Soil

KodaPure · CAT # KBTNA-S-26

IMPORTANT

NOTE BEFORE START

1. Mix all buffers and suspensions thoroughly immediately before use. Reagents such as MB, PW, and LBIII may contain settled or undissolved material; resuspend completely to ensure consistent pipetting and reliable performance.
2. For samples with high protein content, add Proteinase K during the 56 °C incubation step, where applicable after LBII has been added. Use Proteinase K and follow the validated amount for your sample type.
3. To isolate DNA only, treat the sample with RNase A as described in the Notes section.
4. If the LBII bottle shows a substantial reduction in volume before it has been opened, do not use the reagent. Report the issue to KreatBio and request a replacement.

PROTOCOL SUMMARY

SAMPLES	SAMPLE INPUT	INHIBITOR REMOVAL	ELUATE STORAGE
Soil (Not suitable for soil with extremely high inhibitor/minerals.)	**30 mg soil**	**Tube A, 3 min**	**4 °C short / −20 °C long**

REAGENTS SUPPLIED

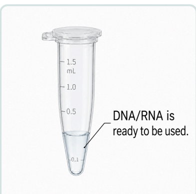
- 1 X 72 mL PW — Pre-Wash Buffer
- 1 X HB01 — Homogenizing Beads 0.1 mm
- 50 X Tube A — Inhibitor-removal tube
- 1 X 0.8 mL MB — Magnetic Beads
- 1 X 72 mL WBII — Wash Buffer II
- 1 X 37 mL LBI — Lysis Buffer I
- 1 X HB05 — Homogenizing Beads 0.5 mm
- 1 X 17 mL BB — Binding Buffer
- 1 X 37 mL WBIII — Wash Buffer III
- 1 X 5 mL EB — Elution Buffer

ADDITIONAL MATERIALS AND APPARATUS REQUIRED

- Centrifuge machine suitable for 1.5 mL microcentrifuge tubes
- Vortex machine or homogenizer
- Magnetic rack
- Tube rack
- 1.5 mL microcentrifuge tubes
- Pipettes and compatible pipette tips

METHOD

1. Add 30 mg soil to a 1.5 mL tube.
2. Mix PW well before use. Add 700 µL PW and mix vigorously by hand. Leave it for 2 min. Centrifuge at max speed (~12,000 × g) for 2 min. Discard the supernatant and repeat this step.
3. Add homogenizing beads HB01 and HB05 to the sample tube up to ~5 mm height. Add 700 µL LBI. Then, vortex the tube at 3000–4000 rpm for 5 min.
4. Centrifuge the tube at max speed for 3-5 min. Then, transfer 400 µL of clear supernatant to a new tube. (Avoid taking the dirty pellet.) Open one Tube A and add all its contents to the supernatant. Mix vigorously by hand and leave for 3 min. Centrifuge for another 5 min.
5. Transfer 300 µL of the treated supernatant into a new tube. Bring MB to room temperature and mix it well before use. Add 300 µL BB and 15 µL MB to the sample tube. Mix gently and leave for 10 min.
6. Place the tube on the magnetic rack until the supernatant is clear (~2 min with KreatBio rack). Carefully decant or pipette out the clear supernatant without disturbing the black beads. Leave a small amount if necessary to reduce bead loss, then remove the tube from the rack.
7. Add 700 µL WBIII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant while still on the magnetic rack. Once done, remove tube from the rack.
8. Add 700 µL WBII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant, remove the tube from the rack, and repeat this step (Step 8). Tip: A quick spin, then back on the rack, removes excess WBII if there is. Remove the tube from the rack.
9. Open the tube lid and leave it for 5 min to air-dry.
10. Add 50–80 µL EB to the black beads in the tube. Mix thoroughly and leave for 10 min. If some beads remain stuck to the tube wall, mix as well as possible and proceed.
11. The DNA/RNA is contained in the eluate. Place the tube on the magnetic rack for approximately 1 min, then transfer only the clear supernatant to a new tube leaving the black beads.
12. Salt or inhibitor carryover can disrupt PCR; if samples fail to amplify, dilute 1:10 or 1:50 and rerun.



TROUBLESHOOTING

EVENT	LIKELY CAUSE	SUGGESTED FIX
Low DNA yield	Too much sample, or poor homogenization	Use less starting material and homogenise fully.
RNA in a DNA-only eluate	RNase A not used	Add RNase A and incubate as noted.
Bead carryover	Beads disturbed during transfer	Leave on the magnet until clear, then transfer carefully.
Brown eluate	Soil, fat, or debris carried over	Repeat extraction and transfer only clear lysate, avoid the pellet and fat layer. Centrifuge if it helps with separation. If this is for PCR amplification, dilute eluate to 1:10 to 1:50.
Poor PCR result	Inhibitor carryover or too much template	Dilute the eluate 1:10 to 1:50 and avoid debris.
Soil inhibition	Humic substances carried over	Use Tube A for inhibitor removal, or dilute the eluate to 1:10 to 1:50 and rerun PCR amplification.

SAFETY

SAFETY & HANDLING

- Research use only. Wear gloves, a lab coat, and eye protection.
- BB, WBII, and WBIII are flammable — keep away from heat and flame.
- LBI and WBIII contain guanidine (GuHCl) — do not mix with bleach.
- Dispose of waste according to local rules.
- TNA means total nucleic acid — these kits recover both DNA and RNA.